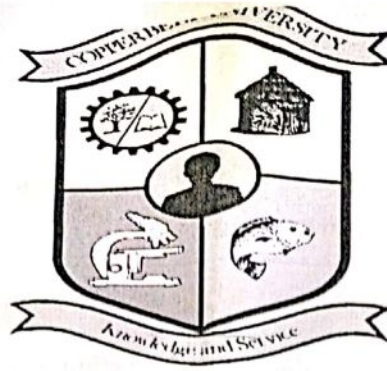




THE COPPERBELT UNIVERSITY

SCHOOL OF MEDICINE

MBS 220

PATHOLOGY

End of Term test

26 Jun 2024

Student Identification Number:-----

Program:.....

Time: 2:00

INSTRUCTIONS

This question paper has 4 sections:

SECTION A: One best answer, 20 Marks)

SECTION B: True/ False 10 marks)

SECTION C: Short note, 30 marks

SECTION D: long essay 40

The Paper has a total mark of 100%

ALL THE BEST

SECTION A – Choose one best answer, one mark for each correct answer, $\frac{1}{4}$ a mark will be deducted for any wrong answer.

1. Which of the following statement is true about inflammation?

- a. It always results into eliminations of the stimuli.
- b. Normally results into tissue repair and always has an advantage to the host.
- c. The whole purpose of it is to deliver defence mechanisms to the site of injury.
- d. Both macrophages and neutrophils come in the second phase of acute inflammation.

2. What is the main difference between acute inflammation and chronic inflammation?

- a. In acute, there is presence of neutrophils
- b. In chronic, there is presence of lymphocytes
- c. In chronic, the response is very specific while in acute its none specific
- d. Acute has 3 phases while chronic has one phase.

3. Which one of the following statements is the best about the factors mediating immune system?

- a. Pathogen associated molecular pallets can be recognised by any cell of innate immunity.
- b. The macrophages and dendritic cells have receptors that recognises pathogen associated molecular pallets.
- c. The interactions between Tall like receptors and macrophages start the reactions of phagocytosis.
- d. The interactions of Tall like receptors and the pathogenic pallet results into multiple chemical mediators.

4. Which one of the cyclooxygenase end products has multiple chemical mediating functions during acute inflammation?

- a. Prostaglandin I₂
- b. Prostaglandin D₂
- c. Prostaglandin E₂
- d. Prostaglandin I₂ and E₂

5. Which of the following products of 5-lipoxygenase also acts as neutrophils chemotaxis?

- a. Leukotriene B₄
- b. Leukotriene C₂
- c. Leukotriene D₄
- d. Leukotriene E₄

6. The following may result in stimulations of chronic inflammations except one;

- a. Chronic bacterial infections like *E. coli* and *Mycobacterium*
- b. Cancers and fungal infections
- c. Fungal and parasitic infections
- d. Genetic disease that results into autoimmune responses.

7. Which of the following is true about pathogen associate molecular pallets associated with MHC II?

- a. It can only be recognised by the CD8 cell.
- b. It results into cytotoxic responses.
- c. Results into immune response activation.
- d. It results into CD4 degradation.

8. Which one of the following statements is true about cascating granuloma?

- a. The denoting cell is epithelioid
- b. Can be caused by cat scratch disease.
- c. It has necrotic debris at the centre
- d. Results from sarcoidosis disease.

9. All of the following may results in delayed wound healing except...

- a. Lack of zinc in the diet
- b. Lack of vitamin C
- c. Infections
- d. Excessive alcohol intake

10. A patient came with a wound measuring 8cm by 12cm, two weeks later, the patient came back with the same wound measuring 3cm by 6cm. Which of the following statements below is correct about the wound mentioned above?

- a. There is too much collagen production that facilitates healing.
- b. The reductions in sizes of the wound is due to actions of collagenase enzymes.
- c. The reductions in the sizes of the wound is due to the actions of fibroblasts.
- d. The whole issue is due to the contractile ability of the myofibroblasts.

11. Regarding the history of pathology

- a. Gabriel Fallopius described human lymphoid tissue
- b. Malpighi is known as 'the father of histology
- c. Rudolf Virchow is known as father of exfoliative cytology
- d. Tijo and Levan described the structure of DNA

12. Which of the following may cause Metaplasia

- a) Denervation,
- b) Diminished blood supply
- c) Inadequate nutrition,
- d) Chronic irritation of the epithelium

13. Which of the following are features of irreversible injury

- a) Nuclear chromatin clamping
- b) Formation of cytoplasmic blebs
- c) Mitochondria swelling
- d) Nuclear pyknosis

14. Coagulative necrosis

- a) Is caused by bacterial infection
- b) Has a cheese-like gross appearance
- c) May be caused by thrombosis
- d) Is a reversible change

15. Which of the following is not an intracellular accumulation

- a) Dystrophic calcification
- b) Metastatic calcification
- c) Fatty change
- d) Iron overload

16. Calcification

- a) Is a characteristic features of metaplasia
- b) Occurs only in dead cells
- c) is an extracellular accumulation
- d) may cause organ dysfunction

17. Mechanisms of cellular Aging

- a) Increasing DNA damage,
- b) Decreased cellular replication
- c) Reduced regenerative capacity of tissue stem cells
- d) Accumulation of metabolic damage

18. Atrophy

- a) Is always a pathological response
- b) Is a decrease in the size of cells
- c) Is characterised by calcification of the organs
- d) Is an increase in the cell size

19. Metaplasia

- a) Is an increase in the number of cells
- b) Is synonymous with cancer
- c) Is an irreversible adaptive response
- d) Arise by genetic reprogramming of stem cell

20. Pathological calcification

- a) Implies the normal deposition of calcium salts in living tissues.
- b) May cause organ dysfunction
- c) Is an adaptive change in labile tissues
- d) Occurs in dead or dying tissues

SECTION B – True/ False

¼ a mark will be deducted for any wrong option, one mark for each correct answer.

1. Analyse the following statements about edema and decide whether they are true or false:-

- a. Accumulation of excess fluid in the extravascular tissues.
- b. Edema is an abnormal response of the body to inflammation or injury.
- c. Exit of fluid from arteriolar end is balanced by inflow at venular end of microcirculation.
- d. Small residual amount of excess interstitial fluid is drained by venular end of microcirculation.

e. immunoglobulins contribute to the colloid osmotic pressure (COP) of blood.

2. Analyse the following statements and decide whether they are true or false:-

- a. Osmosis takes place at the arterial end of the capillary.
- b. Gases are exchanged by diffusion.
- c. Filtration takes place at the arterial end of the capillary.
- d. Osmosis - a process by which molecules of a solvent tend to pass through a semipermeable membrane from a more concentrated solution into a less concentrated one.

e. Increased hydrostatic pressure can contribute to the formation of edema.

3. Analyse the following statements and indicate whether they are true or false:

- a. Reduced plasma oncotic pressure can be a cause of edema.
- b. Obstruction of fluid clearance in the lymphatic system contributes to the formation of edema.
- c. Inflammatory edema is caused by increased hydrostatic pressure at the arterial end of the capillary.
- d. Sodium and water retention plays an important role in pathogenesis of edema.
- e. Hypovolemia - an increased volume of circulating blood in the body.

4. Analyse the following statements about pulmonary edema and indicate whether they are true or false:

- a. It is the accumulation of fluids in the lungs due to the blockage of the pulmonary veins.
- b. The pleural cavity can also be filled with fluid.
- c. Acute myocarditis (inflammation of the myocardium) is one of the causes of pulmonary edema.
- d. Pink frothy sputum is a symptom of edema.
- e. Edema of the lungs is one of the symptoms of acute respiratory distress syndrome.

5. Analyse the following statements about Hyperemia and indicate whether they are true or false:

- a. Hyperemia is an excess of Blood in an Organ.
- b. Active hyperemia is caused by impaired exit of blood through venous pathways.
- c. Passive hyperemia or congestion is usually a physiologic response to increased functional demand.
- d. Neurogenic and hormonal influences play a role in passive hyperemia.
- e. The impaired venous drainage is called venous congestion or passive hyperemia.

6. Analyse the following statements about chronic venous congestion (CVC) and indicate whether they are true or false:

- a. It may be local or systemic.
- b. The affected tissue or organ is red in colour.
- c. Systemic venous congestion results from right-sided heart failure.
- d. Pulmonary congestion is due to mechanical overload in aortic stenosis.
- e. Cirrhosis of the liver causes a local venous congestion.

7. Analyse the following statements about the morphology of chronic venous congestion (CVC) and indicate whether they are true or false:

- a. Heart failure cells (i.e. hemosiderin-laden macrophages) are present within alveolar spaces of the lungs.
- b. The alveolar septa within lungs are widened and thickened.
- c. The spleen shows nutmeg appearance.
- d. The liver shows brown induration.
- e. Spleen shows dilated sinuses and presence of Gamma-Gandy body.

8. Analyse the following statements about consequences of chronic congestive heart failure and indicate whether they are true or false:

- a. Pulmonary hypertension.
- b. Acute peritonitis (i.e. accumulation inflammatory exudate within abdominal cavity).
- c. Enlargement of the spleen.
- d. Pleural effusion.
- e. Pitting edema of the leg.

9. Analyse the following statements about Hemorrhage and indicate whether they are true or false:

- a. Hematoma is the extravascular accumulation of blood clot e.g. into the tissues.
- b. Hemopericardium is hemorrhage into the pericardial sac.
- c. Hemothorax is accumulation of blood in the pleural cavity.
- d. Hemorrhage also results from damage to lymphatic capillaries.
- e. Vitamin C deficiency is associated with capillary fragility and bleeding.

10. Analyse the following statements about thrombosis and indicate whether they are true or false:

- a. Thrombosis is the process of formation of solid mass in circulation from the constituents of flowing blood.
- b. Thrombi decrease or stop the blood supply to parts of an organ or tissue.
- c. Local venous congestion results from intraluminal occlusion by thrombosis.
- d. A thrombus is an aggregate of coagulated blood containing platelets, fibrin and entrapped cellular elements, within or out of a vascular lumen.
- e. Formation of a thrombus is thrombosis.

SECTION C – Answer two in each part, 5 marks each.

PART A

1. Discuss the Methods used in the study of pathology
2. Discuss the 7 causes of cell injury
3. Discuss liquefactive necrosis

PART B

1. Steps involved in transformations of Macrophages to Epithelioid.
2. Types of granulomatous.
3. Activations of CD8 cytotoxic cells.

PART C

1. Discuss gas/air embolism.
2. Discuss arterial thromboembolism.

SECTION D – Answer one in each part, 13 marks each.

PART A

1. Discuss in details would healing by repair.
2. Discuss in details the mediators of acute inflammations.

PART B

1. Discuss in detail 2 intracellular and 2 extracellular degenerative changes in tissues
2. Discuss in detail the 6 molecular mechanisms and biochemical pathway of cell injury

PART C

1. Discuss Sequelae of Thrombosis (in details).